MIL-I-4430
28 March 1985
SUPERSEDING
MIL-I-4430
26 December 1973

MILITARY SPECIFICATION

INDICATOR, CONNECTING ROD ALIGNMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an indicator for checking alignment characteristics of piston connecting rods used in internal combustion engines.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more indicators for first article inspection (see 4.3). The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The indicator shall be of the manufacturer's latest design which shall meet the performance requirements specified herein. The indicator shall be designed to check the alignment of connecting rod assemblies, and to indicate the extent and direction of rod deformations and of misalignments

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caused by bends, twists, offsets, and lack of squareness. The connecting rod indicator shall check connecting rods with or without the piston assembled to the rod, and shall be designed for use on a bench or tabletop. The design shall be such that all parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the automotive equipment industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction.

* 3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

* 3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with approved engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.4.5 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (of a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

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* 3.4.6 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.5 Performance and product characteristics. The indicator shall be capable of checking connecting rod assemblies to determine the extent and direction of distortion as a result of bends, twists, offsets between the wrist pin and crankshaft journal ends, and lack of squareness of the wrist pin hole with the wall of the piston. The indicator shall be capable of handling connecting rod assemblies, within the range specified in 3.5.1, either with or without piston.

3.5.1 Capacity. The indicator shall check alignment of rods within a minimum center-to-center (crankshaft to piston pin) length range of 3.15625 through 15.50 inches, involving piston pin bearing diameters within a minimum range of 0.65625 through 2.50 inches, crankshaft bearing diameters with a minimum range of 1.50 inches through 4.25 inches, and piston diameters of 2.25 through 7.00 inches (see 4.6.2).

3.5.2 Accuracy. The indicator shall possess an alignment accuracy such that it shall measure rod offset within 0.0005 inch, and shall measure a deviation of parallelism of the axis of the connecting rod bearings within 0.0003 inch per inch of bearing axis length in bend and twists (see 4.6.1).

3.6 Components.

3.6.1 Mounting base. The mounting base shall be an iron or steel casting or forging. The mounting base shall include holding devices to position and retain, without straining or damaging, the connecting rod assemblies in 3.5.1. The mounting base shall be such as to allow one person to do the checking, using the gages and devices in 3.6.2, without elaborate setup and removal procedures. Means shall be provided for attaching the base to work bench or tabletop.

3.6.2 Gages and checking devices. The connecting rod assemblies shall be checked by measuring gages or checking devices which are separate from, or which form an integral part of, the mounting base specified in 3.6.1. The gaging and measuring devices shall provide the accuracy specified in 3.5.2, and shall reflect direct indications or readings requiring no conversion process to acquire results.

* 3.6.3 Carrying case. The indicator shall be provided with a carrying case with compartments and padded cushioning for the indicator. The fully loaded and closed case shall retain its position, shape, and security during military use and handling.

3.7 Tools. Unless otherwise specified (see 6.2.1), the following tools shall be furnished with each indicator. The tools shall have hardened points and shall be designed to fit all types of piston pins.

1 tool, bending
1 tool, twisting

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3.8 Product identification. The indicator shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall assure that the indicators shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Accuracy test (see 4.6.1).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105. For product examination (see 4.5), general inspection level II and an Acceptable Quality Level (AQL) of 1.0 shall be used. For the accuracy test (see 4.6.1), special level S-3 with no defects allowed shall be used. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 shall be used.

* 4.5 Product examination. The indicators shall be examined to determine compliance with requirements of Section 3 and the contract. As a minimum, the following characteristics shall be examined:

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CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design	3.2
Material	3.3
Reclaimed materials when required	3.3.1
Construction	3.4
Castings and forgings	3.4.1
Welding, brazing, and soldering	3.4.2
Fastening devices	3.4.3
Threads	3.4.4
Painting and finishing	3.4.5
Interchangeability	3.4.6
Components	3.6
Mounting base	3.6.1
Gages and checking devices	3.6.2
Carrying case	3.6.3
Tools	3.7
Product identification	3.8
Workmanship	3.9

A certificate of conformance may be provided to satisfy the examination requirements when more than a visual or dimensional verification would be necessary.

4.6 Tests.

* 4.6.1 Accuracy test. Obtain a distorted connecting rod assembly having a piston 6 to 7 inches in diameter, a crankshaft bearing diameter of 3.75 to 4.50 inches, and a center-to-center crankshaft to piston pin length of 14.00 to 15.50 inches. The connecting rod assembly, containing bends, twists, and offset, shall be checked by standard laboratory measuring methods to determine if the misalignments exceed the following:

(a) Rod offset (bends & twists) by more than 0.0005 inch.
(b) Parallelism of the axis of the connecting rod bearings with the axis of the piston pin ends by more than 0.0003 inch.
(c) Squareness of the wrist pin hole with the wall of the piston.

The connecting rod of the connecting rod assembly shall be checked on the indicator and the indicator shall show the connecting rod to be misaligned. The connecting rod shall be checked both with and without the piston assembled to the connecting rod. The squareness of the wrist pin hole with the wall of the piston shall also be checked on the indicator. Squareness shall be checked both with and without the piston assembled to the connecting rod. The connecting rod shall be subjected to the straightening process, using the accessory bending and twisting tools, and then rechecked on the indicator until the indicator shows the connecting rod to be in correct alignment. The straightened connecting rod shall be checked by standard laboratory measuring methods to determine if the indicator conforms with the accuracy requirements of 3.5.2

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* 4.6.2 Capacity test. The accuracy test of 4.6.1 shall be repeated using a connecting rod assembly with a piston 2.25 to 3.00 inches in diameter, a crankshaft bearing diameter of 1.50 to 2.25 inches, and a center-to-center crankshaft to piston pin length of 3.15625 to 5.00 inches.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When military (Level A) preservation is required, it shall be so specified in the contract.).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The indicator shall be cleaned and dried in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with process C-1. Drying shall be accomplished in accordance with procedure D-1 or D-4, or a combination of both procedures.

5.1.1.2 Preservative application. Immediately after cleaning and drying, all bare metal surfaces other than stainless steel, not painted or plated, shall be coated with type P-7 preservative of MIL-P-116. Preserved areas shall be wrapped with barrier material conforming to grade A of MIL-B-121 The barrier material shall be secured with tape conforming to PPP-T-60, type optional.

5.1.1.3 Unit protection. The method of unit packaging shall be in accordance with method I of MIL-P-116. Technical data shall be preserved using method IC-1 of MIL-P-116. All components comprising one indicator, including any technical data, shall be placed within the cushioned carrying case. Cushioning material that is noncorrosive to the items packaged shall be used to prevent movement or damage. After cushioning to prevent movement, the carrying case shall be placed in a snug fitting fiberboard box conforming to PPP-B-636, type CF, class WR. Closure of the box shall be in accordance with PPP-B-636.

* 5.1.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

* 5.2 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

5.2.1 Level A. The indicators, preserved as specified in 5.1.1, shall be packed in a snug fitting container conforming to PPP-B-601 (overseas type) or PPP-B-621, class 2. The quantity packed shall be governed by the size and weight limitations of the container specification. Closure and strapping shall be in accordance with the applicable container specification.

* 5.2.2 Level B. The indicators shall be packed as specified in 5.2.1 except that the containers shall conform to PPP-B-601, domestic type, or PPP-B-621, class 1.

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* 5.2.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.3 Marking. Marking shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military (Levels A and B) marking is required, it shall be so specified in the contract.

* 5.3.1 Level A and B marking. Level A and B marking shall be as specified in MIL-STD-129.

* 5.3.2 Industrial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. This specification covers an alignment indicator for checking all types of connecting rods, with or without pistons assembled. The indicator is also used to check squareness of the piston pin bearing with the outside surface of the piston.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Painting if different (see 3.4.5).
(d) Tools if different (see 3.7).
(e) Product marking if different (see 3.8).
(f) First article inspection if different (see 4.3).
(g) Quality conformance inspection if different (see 4.4).
(h) Sampling when required (see 4.4.1).
(i) Level of preservation, packing, and marking if different (see 5.1, 5.2, and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Preparing activity:
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1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
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5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
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